# Lecture 1: Introduction to computer architecture

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## Learning Goals

- Figure out who the man at the front of the room is.
- Understand the administrative parts of computer architecture within ENCE260.
- Have a brief, conceptual understanding of computer components as will be explored within ENCE260.
- Understand logic statements, variables, and values.

## Further Reading

ENCE361 - Next year
ENEL373 - Next year

## References

Rizzoni and Kearns. *Principals and Applications of Electrical Engineering*, Chapters 13 & 14.

Version: 2.0

# 1 Introduction — About me

Kia ora, my name is Lui Holder-Pearson and I will be lecturing for computer architecture.

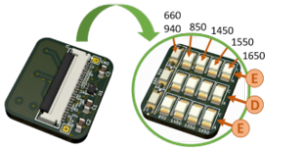
You can find me:


- Email: lui.holder-pearson@canterbury.ac.nz
- Office: A509, fifth floor of the Link Building

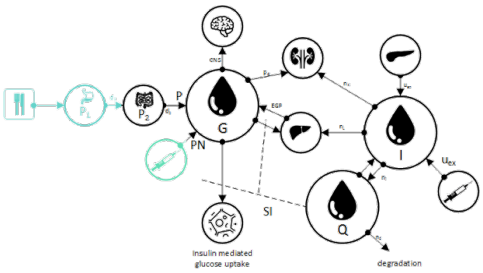

## Research

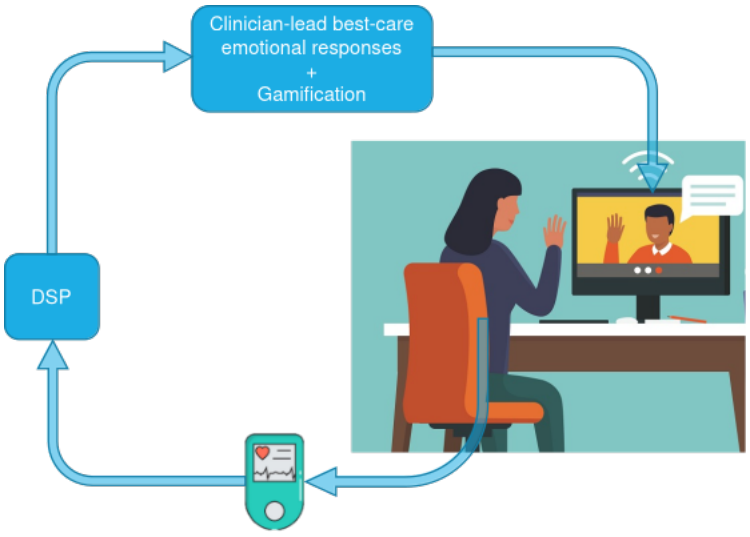
Medical devices, specifically making them cheaper.

- Diabetes technologies — sensors and pumps
- Modelling for control
- Virtualised therapy for mental health conditions





# 2 Admin

## Lecture plan Version 1.0, updated 8/07/2026

| Week | Lectures | | | Other | |
|---|---|---|---|---|---|
| | Wednesday 11am | Thursday 1pm | Friday 9am | Tutorial | Quiz |
| 1 | 1: Introduction | 2: Data representation | 3: Memory and buses | | |
| 2 | | 4: Boolean logic | | Tutorial 1 | Quiz 1 |
| 3 | | 5: CPU | | | |
| 4 | | 6: Combinational logic | | | |
| 5 | | 7: GPIO | | Tutorial 2 | Quiz 2 |
| 6 | | 8: Sequential logic | | | |
| 7 | | *Reserved for Embedded Systems* | | | |
| 8 | | 9: Peripherals | | *Embedded Systems* | |
| 9 | | 10: Execution | | *Embedded Systems* | |
| 10 | | 11: Function calls | | Tutorial 3 | Quiz 3 |
| 11 | | | | *Embedded Systems* | |
| 12 | | 12: Summary/recap | | Exam tutorial | Quiz 4 |

This plan is subject to change. Changes will be announced in lecture and on Learn.

## Assessment (Computer-architecture specific ONLY)

1. Quizzes (4x ≈1%). Quizzes are due 6pm on the Friday of weeks indicated.
2. Half of the final examination (50%): Shared with embedded systems.

## Who to contact

- For questions around the computer architecture *content*, please use the forum on Learn.
- For queries pertaining specifically to computer architecture *administration*, contact me, Lui: lui.holder-pearson@canterbury.ac.nz. I am also the course coordinator for ENCE260, so if you are missing assignments or other administrative queries this should go to Richard.
- If your issue is relating to assessment or lecture material, please in the first instance contact your class representatives, as they have the proper means to communicate with faculty staff.
- For queries around the C programming aspect, contact Dr Andrew Bainbridge-Smith.
- For queries around the embedded systems content, contact (probably by now Dr) Sam Spekreijse.

# 3 Computers — What is ENCE260?

## ENCE260 — What is a computer?

How many computers do you own? How many do you have on your person at the moment?

Computers are increasingly ubiquitous: they are being *embedded* inside more and more devices. These computer *systems* that are *embedded*, are differentiated from general computing devices that we are used to using on a daily basis:

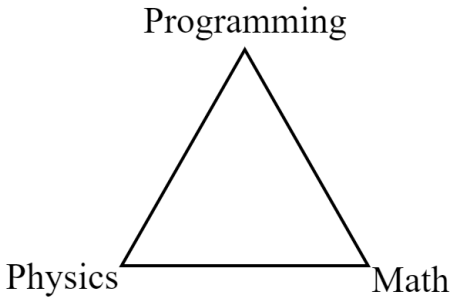
Dfn: **Embedded system**: simply a small computer that has a specific task.

| **Embedded system** | **Personal computer** |
|---|---|
| Specific task | General task |
| Usually autonomous | Requires real-time human input |
| Low computing power | High computing power |
| Within other systems | Standalone |
| Requires no specific UI, if any | Requires large amount of UI to function |

For ENCE260 we will be looking at the specific architecture of embedded systems, in particular the ATMega32u2, which will be used for the assignment in term 4.

Just as architecture in a traditional sense pertains to the design, layout, and interface of different spaces (**components**), and how those spaces are related to each other (**connections**), and what the underlying **material** and **functionality** of them is.

### 3.1 Computer architecture in two pictures:



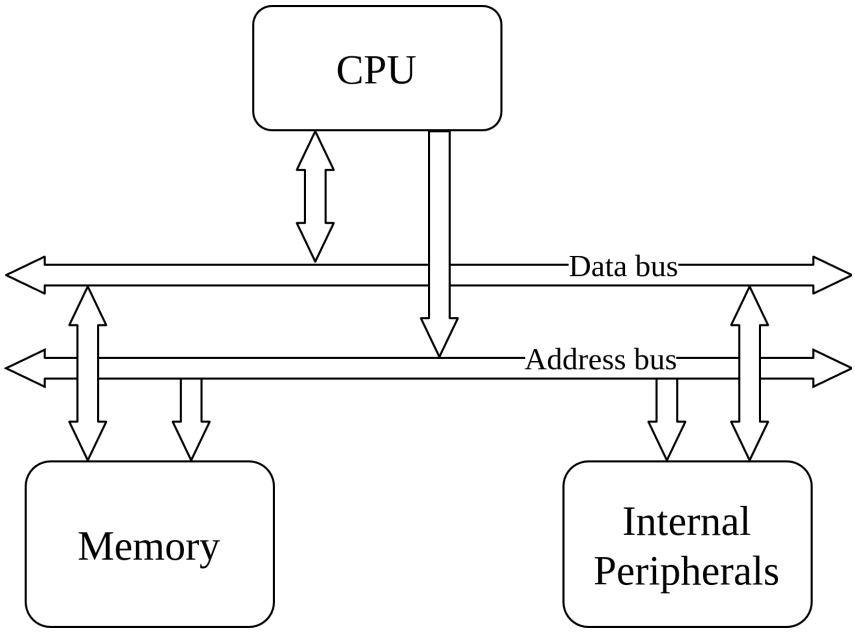




Figure 2: A generic embedded system architecture.

Over the course of the next 11 lectures we will look at computer architecture, showing the interface between software (Dr Bainbridge-Smith's C component), and having hardware which achieves a purpose (Dr Richard Clare's Embedded systems component.

# 4 Boolean/Digital logic

*Boolean* is the term used to explicitly describe that data type in computer science that is able to take only one of two values, where *digital* refers to information being represented by physical quantities which may only take two states. For all intents and purposes, they are the same thing in ENCE260.

Digital logic is the branch of mathematics that we use to manipulate binary maths. We need to understand it because it allows us to handle specific kinds of *information* that we use in computers.

In the world around us, we are very used to dealing with analogue/continuous information. For example:

- Current position.
- Voltage.
- Temperature of a room.
- Length of a piece of string.

However, electrical memory is only able to store one of two states, and therefore we need to be able to represent information in this digital, *yes*/*no* manner:

- Am I in E8 lecture theatre?
- Is the voltage 0.0 $V$?
- Am I warm enough?
- Is there a piece of string?

The answers to these questions — *logical values* — are represented in several different, but exactly equivalent ways:

| **Representation** | | |
|---:|:---:|:---:|
| Numerical value: | 0 | 1 |
| Logical value: | `FALSE` | `TRUE` |
| Voltage: | Low | High |
| Presence: | Absent | Present |
| Switch state: | Open | Closed. |

## Summary

- The man at the front of the room is Lui, who will be lecturing the computer architecture aspect of ENCE260.
- There are 11 lectures, 6 tutorials, 4 quizzes, and a portion of the exam throughout the semester.
- The three main components of a computer are:
  1. Memory and Buses
  2. Central processing unit
  3. Internal peripherals
- Understand logic statements, variables, and values:
  → Logic statements are sentences that answer a yes/no question.
  → Logic variables are the programming variable names in which we store the answers.
  → Logic values are the values stored within the above variables: either 0/1, `FALSE`/`TRUE`, or the physical representation of how this electrically achieved.

## Next Lecture: Lecture 2 - Data representation

# 5 Questions?